

Pharos University in Alexandria
Faculty of Dentistry
Department of Restorative Dentistry
Student Code No. 202203112

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Name of Candidate

Mohamed Talaat Mohamed AbdelMoaty ElAbd

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**COMPARATIVE STUDY OF WEAR RESISTANCE
AND SURFACE ROUGHNESS OF INJECTABLE VERSUS
CONVENTIONAL COMPOSITE RESIN — IN VITRO STUDY**

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Supervision Committee

1. Prof. Wegdan M. Abdel-Fattah
2. Asst. Prof. Emad M. El-Sayed (Main supervisor)

ROLE OF SUPERVISORS

1. Prof. Wegdan M. Abdel-Fattah

Conceptualization/formulation of study plan and design of the study.

Analysis and interpretation of study results.

Supervision of study execution and writing.

Guidance of paper/thesis writing.

2. Asst. Prof. Emad M. El-Sayed

Supervision of the clinical/experimental procedures.

Conceptualization/formulation of study plan and design of the study.

Supervision of study execution and writing.

Revising the thesis and/or manuscript.

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LIST OF ABBREVIATIONS

ANOVA — analysis of variance

CAD/CAM — computer-aided design / computer-aided manufacture

ISO — International Organization for Standardization

LED — light-emitting diode

Ra — arithmetic mean roughness

RDA — relative dentine abrasivity

SD — standard deviation

S-PRG — surface pre-reacted glass-ionomer

vol% — percentage by volume

Δ Ra — change in Ra (after – before)

ABSTRACT

Background. Highly filled injectable composites are marketed for use in load-bearing restorations. Independent data comparing their gravimetric wear and surface roughness after standardised toothbrush abrasion with those of a conventional nanohybrid resin remain limited.

Aim. To compare percentage weight loss and arithmetic mean roughness (Ra) of Beautifil Flow Plus X F00, G-aenial Universal Injectable and Beautifil II LS after 10 000 toothbrushing cycles.

Methods. Thirty-six discs (10 mm diameter $\times$ 1 mm thick; n = 12 per group) were subjected to 10 000 brushing cycles at a load of 2 N in a Colgate Total slurry prepared according to ISO guidance. Specimen mass was recorded before and after abrasion. Ra was measured by contact profilometry. Percentage weight loss was compared by one-way ANOVA. Ra after brushing was compared by the Kruskal–Wallis test ($\alpha = 0.05$).

Results. Mean percentage weight loss was 1.49 ± 3.14 % for Beautifil Flow Plus X F00, 0.65 ± 0.37 % for G-aenial Universal Injectable and 0.41 ± 0.25 % for Beautifil II LS ($p = 0.329$). Mean Ra after brushing was 0.104 ± 0.017 μm , 0.100 ± 0.023 μm and 0.194 ± 0.050 μm respectively ($p < 0.001$). Each injectable resin was significantly smoother than Beautifil II LS. The two injectable materials did not differ from each other. One Beautifil Flow Plus X F00 disc lost approximately 11.4 % of its mass and accounts for the large standard deviation in that group.

Conclusions. Under this toothbrushing protocol, gravimetric wear did not differ significantly among the three resins. Both injectable materials remained significantly smoother than the conventional nanohybrid. Mean Ra of the injectable resins stayed below the 0.2 μm plaque-retention threshold, whereas the nanohybrid mean lay on that threshold.

Keywords. Injectable composite; nanohybrid composite; toothbrush abrasion; wear; surface roughness; giomer.

CHAPTER 1

INTRODUCTION

Resin-based composites have become the principal material for direct restoration of both anterior and posterior teeth. Their clinical performance is determined not only by handling and aesthetics but by the behaviour of the finished surface under the conditions of the mouth. Two surface properties are of particular concern. Progressive wear flattens occlusal anatomy, may reduce vertical dimension and can disturb the occlusal scheme. Increased roughness, independently of the volume of material lost, retains plaque, takes stain and favours secondary caries at the restoration margin ^(1,2). Daily toothbrushing contributes to both processes. The abrasive particles of a dentifrice act as a third body between the bristle and the restoration: the resin-rich surface layer is removed first, filler particles are then left standing proud or are plucked from the matrix, mass is lost and the arithmetic mean roughness (Ra) rises ^(3,4).

For many years the materials used in load-bearing cavities were high-viscosity pastes, while low-filled flowable resins were reserved for liners, small defects and situations in which adaptation was more important than strength. First-generation flowables typically contained 40–55 wt% filler and showed higher polymerisation shrinkage and poorer wear resistance than their packable counterparts ⁽⁵⁾. A later generation of highly filled injectable composites was formulated to close that gap. By combining a higher inorganic load, a smaller mean particle size and improved silane treatment, manufacturers produced pastes that can still be injected from a syringe, often as a single increment, yet are offered for use in Class I and II restorations ⁽⁶⁾. Two such products, G-aenial Universal Injectable (GC Corporation, Tokyo, Japan) and Beautifil Flow Plus X F00 (Shofu Inc., Kyoto, Japan), are now in routine clinical use. Both were tested in shade A2. Beautifil II LS (Shofu Inc.), a

sculptable low-shrinkage nanohybrid giomer in the same shade, remains a representative conventional paste and was chosen as the control.

The three products were not chosen at random. G-ænial Universal Injectable is a nanofilled dimethacrylate paste with barium glass and silica. Beautifil Flow Plus X F00 is a zero-flow giomer injectable. Beautifil II LS is a packable giomer from the same manufacturer as the Shofu injectable. The comparison therefore asks two separate questions: whether two current injectables of different filler chemistry behave alike under the toothbrush, and whether either of them is at a disadvantage beside a conventional nanohybrid of higher filler volume.

The laboratory literature on these products is uneven. Two-body chewing-simulator studies, in which a steatite antagonist is loaded at about 49 N, have reported favourable volumetric wear and flexural strength for highly filled injectables ⁽⁶⁾. Those data describe occlusal-contact wear. They do not describe the three-body abrasion produced by a toothbrush and a dentifrice slurry, which is the wear mode that acts on free cleanable surfaces — including the labial and palatal faces of anterior restorations — and on margins throughout the life of a restoration. Toothbrushing tests conducted according to ISO/TR 14569-1, at a load of 2 N and for 10 000 strokes, are conventionally treated as the laboratory equivalent of one year of twice-daily brushing ^(3,7,8). Direct comparison of Beautifil Flow Plus X F00, G-ænial Universal Injectable and Beautifil II LS under that protocol has not been published.

The clinical meaning of the roughness values obtained after such a test is judged against two well-established thresholds. Bollen, Lambrechts and Quirynen concluded that plaque retention increases once Ra exceeds approximately 0.2 µm and that further smoothing below this value produces no additional clinical benefit ⁽¹⁾. Jones, Billington and Pearson later showed that volunteers can detect roughness with the tip of the tongue once Ra lies between 0.25 µm and 0.50 µm, and they advised finishing restorations to no more than 0.50

$\mu\text{m}^{(2)}$. These two figures provide the frame within which the roughness results of the present study are interpreted.

A controlled in-vitro comparison of gravimetric wear and Ra after standardised toothbrushing is therefore required if the clinical claims made for highly filled injectable resins are to be examined against a conventional nanohybrid paste under identical abrasive conditions.

1.1 Aim of the study

To compare wear resistance, expressed as percentage weight loss, and surface roughness (Ra) of two highly filled injectable composites with a conventional nanohybrid composite after standardised toothbrushing.

1.2 Objectives

1. To measure specimen mass before and after 10 000 brushing cycles and to calculate percentage and absolute weight loss.
2. To measure Ra before and after the same abrasion and to calculate ΔRa .
3. To compare the three materials using the statistical tests appropriate to each outcome.

1.3 Null hypotheses

1. There is no significant difference in percentage weight loss among Beautifil Flow Plus X F00, G-aenial Universal Injectable and Beautifil II LS.
2. There is no significant difference in Ra after toothbrushing among the same three materials.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Wear of restorative resins

Wear is the progressive loss of substance from two surfaces that are in relative motion ⁽⁹⁾. In the mouth the process is rarely of a single type. Attrition occurs at sites of direct occlusal contact and is essentially two-body wear. Abrasion at non-contact sites is usually three-body wear, with food particles or dentifrice abrasive as the loose third body. Corrosion, produced by chemical softening of the resin, and fatigue, produced by cyclic loading, run in parallel with both mechanical modes ^(9,10). Laboratory tests isolate one or two of these mechanisms in order to rank materials. They do not reproduce the multifactorial environment of the mouth and therefore cannot be read as a clinical lifetime ⁽¹⁰⁾.

The wear resistance of a resin composite is governed by the filler, the matrix and the interface between them. Filler volume fraction, particle size and shape, inter-particle spacing, the degree of conversion of the resin and the quality of the silane coupling layer all contribute ^(11,12). Condon and Ferracane examined these variables systematically in an oral-wear simulator. Wear increased once filler volume fell below 48 vol% and rose in a linear fashion as the proportion of silane-treated filler was reduced. Raising the degree of conversion reduced wear, although the effect was smaller than that of filler volume or silanation ⁽¹²⁾. Turssi, Ferracane and Vogel later showed, in a set of experimental composites, that reducing filler size improved three-body abrasion resistance for both spherical and irregular particles, provided that conversion was not compromised ⁽²⁴⁾. The practical consequence of this work is that a well-coupled bed of small particles protects the resin from the abrasive and limits the size of the pits left when a particle is lost.

Classical flowable resins, filled at approximately 40–55 wt%, wear more rapidly than packable hybrids for precisely these reasons ^(5,13). The more recent injectable products raise

filler load and reduce particle size in an attempt to retain syringeable handling without accepting that penalty. G-aenial Universal Injectable is listed at 50 vol%, a volume fraction that sits just above the 48 vol% step identified by Condon and Ferracane ^(6,12). Beautifil II LS is listed at 68 vol% ⁽¹⁹⁾.

Laboratory evaluation of toothbrush wear follows ISO/TR 14569-1 ⁽³⁾. The stated scope of that technical report is the wear of artificial teeth and veneering materials, but the recommended brush load (0.5–2.5 N) and the requirement that the dentifrice comply with ISO 11609 have been adopted widely for the screening of restorative composites ^(3,14). Sexson and Phillips estimated that a patient who makes about fifteen strokes per surface, twice daily, produces of the order of 10 000 strokes in a year ⁽⁷⁾. Most subsequent toothbrushing papers have used that figure, or a simple multiple of it, as the laboratory equivalent of one year of hygiene ⁽⁸⁾. The conversion is an estimate; brushing habits vary, and the laboratory brush does not reproduce the anatomy of a restoration. It remains, nevertheless, the convention against which the present cycle count is set.

Wear may be quantified as lost volume, as profilometric depth or as lost mass ^(10,15). Gravimetric loss is simple and repeatable when the balance reads to 0.1 mg and the specimen is dried in a consistent way. It does not describe the topography of the worn surface, and it cannot separate true abrasive loss from a change in water content if the drying step is not the same before and after the test. Volumetric loss from a three-dimensional scan does describe the worn anatomy, but requires equipment that was not used in this study. Heintze and colleagues compared three laboratory methods of wear quantification and showed that they do not always rank materials in the same order ⁽¹⁵⁾. The choice of a gravimetric outcome in the present work is therefore a methodological decision, not a claim that mass is equivalent to worn anatomy.

2.2 Surface roughness

The arithmetic mean roughness, Ra, is defined in ISO 4287 as the arithmetic mean of the absolute ordinate values of the roughness profile within the sampling length ⁽¹⁶⁾. It is insensitive to the shape of individual peaks and valleys, but it is the parameter reported in almost every dental investigation of restoration surfaces and is therefore the value against which clinical thresholds have been set. Contact stylus profilometry remains a standard laboratory method. Optical techniques generate a three-dimensional map and avoid stylus damage, but they are not required in order to rank materials on Ra ^(4,16).

The clinical importance of Ra was established in two independent lines of work. Quirynen and Bollen showed that surface roughness and surface-free energy together determine the amount of plaque that accumulates on restorations ⁽¹⁷⁾. Bollen, Lambrechts and Quirynen then reviewed the available evidence and proposed 0.2 μm as the threshold above which bacterial adhesion increases to a clinically relevant degree; surfaces smoother than this showed no further reduction in plaque ⁽¹⁾. Jones, Billington and Pearson asked volunteers to rank composite discs of known roughness with the tip of the tongue. Discrimination lay between 0.25 μm and 0.50 μm , and the authors recommended that restorations be finished to no more than 0.50 μm if they are not to be felt ⁽²⁾.

Toothbrushing raises Ra. Heintze, Forjanic, Ohmiti and Rousson subjected nine composites and two ceramics to simulated brushing at three loads and showed that the increase in roughness and the loss of gloss depended on the material, the load and the duration of brushing. Hybrid and microhybrid pastes deteriorated more with increasing load than did microfilled resins ⁽⁴⁾. Nanofilled materials, whose particles are small enough to wear with the matrix, generally retain a smoother surface than nanohybrids, in which larger fillers are uncovered or lost ^(8,18).

The dentifrice is an independent variable. da Costa, Adams-Belusko, Riley and Ferracane brushed four composites with Colgate Total (relative dentine abrasivity 70) and with two more abrasive Colgate pastes (RDA 145 and 200). All pastes reduced gloss and

increased Ra; the least abrasive paste produced the smallest change, and composites with smaller fillers were less affected than those with larger fillers ⁽²⁵⁾. Monteiro and Spohr, using a different set of dentifrices, also recorded an increase in Ra after simulated brushing but found that the manufacturer's RDA value did not predict the roughness result in a simple way ⁽⁸⁾. A single dentifrice was therefore used in the present study, and the results apply to that paste.

How the surface is finished before the first measurement also matters. A Mylar-covered, as-cured face is resin-rich and is not the surface that a dentist leaves after rotary finishing and polishing. AlAli, Silikas and Satterthwaite polished three nanohybrid pastes and then brushed them at intervals up to 20 000 cycles. Ra rose and gloss fell at every interval; after 10 000 cycles the polished nanohybrids were already rougher than they had been at baseline, and by 20 000 cycles mean Ra lay between 0.41 μm and 0.49 μm ⁽²⁶⁾. Their baseline values (0.14–0.22 μm) were those of a polished paste, not of a Mylar film. The present discs were left as cured against Mylar. That choice follows the approved protocol. It means that the baseline Ra reported in Chapter 4 describes the as-cured film, and that any comparison with a polished-specimen paper has to be made with that difference in mind. Gloss was not measured here.

2.3 Injectable composite resins

The first generation of flowable composites was introduced to improve adaptation in narrow cavities and to serve as liners. Their filler content was deliberately kept low in order to reduce viscosity, and their mechanical properties were correspondingly inferior to those of contemporaneous hybrid pastes ^(5,13). Baroudi and Rodrigues, in a systematic review, concluded that conventional flowables should not be used as sole restoratives in load-bearing situations ⁽¹³⁾.

Highly filled injectable resins are a later development. G-aenial Universal Injectable contains barium glass and silica at a filler load of 50 vol% in a dimethacrylate matrix ^(6,18,19).

Beautifil Flow Plus X F00 is a zero-flow giomer injectable that incorporates nano-sized surface pre-reacted glass-ionomer (S-PRG) filler ⁽¹⁹⁾. It should not be confused with Beautifil Flow Plus F00, an earlier product that appears in several laboratory papers ⁽⁶⁾. The distinction matters when results are compared.

Syringeability and wear are not the same property. Imai and colleagues measured extrusion force and thread formation together with flexural strength and sliding-impact wear in a set of newer flowable resins. Some of those flowables wore as little as, or less than, the conventional pastes in the same study. Handling did not predict wear: a paste that left the syringe easily was not, on that account, the weaker material under the antagonist ⁽²⁷⁾. The finding supports the manufacturer's claim that a highly filled injectable can be used where a packable paste would otherwise be chosen. It does not, on its own, describe toothbrush abrasion.

Rajabi, Denny, Karagiannopoulos and Petridis compared G-ænial Universal Injectable and Beautifil Flow Plus F00 with a conventional flowable (Tetric EvoFlow) and a nanohybrid paste (IPS Empress Direct) in two-body wear. Specimens were cycled 200 000 times against a steatite antagonist at 49 N under dry conditions, a regimen intended to represent approximately one year of occlusal contact. Both injectables lost less volume and showed higher flexural strength than the paste and the classical flowable; the two injectables did not differ from each other ⁽⁶⁾. The finding is favourable to highly filled injectables, but the wear mode is two-body contact without toothpaste. It cannot be transferred to a toothbrushing assay.

Hasan, Eltoukhy and Zaghloul examined surface roughness after simulated toothbrushing of G-ænial Universal Injectable, a nanofilled paste (Filtek Z350 XT) and a nanohybrid (Tetric N-Ceram). Brushing times of 30 and 60 minutes were taken to represent six months and one year of daily hygiene. Ra increased in all groups. The injectable recorded the lowest mean roughness and the nanohybrid the highest; the two nanofilled resins did not

differ significantly ⁽¹⁸⁾. That design is closer to the present study, although the nanohybrid control and the injectable giomer were not the products tested here.

Tzimas, Pappa, Fostiropoulou, Papazoglou and Rahiotis reviewed highly filled flowable resins used as sole restoratives. They concluded that optical properties are generally adequate, that mechanical properties remain inferior to those of medium-viscosity composites, and that laboratory drawbacks may contraindicate use in extensive cavities, load-bearing areas and patients with parafunction, despite a small number of randomised trials that reported acceptable early performance in Class I and II restorations ⁽²⁰⁾. Material-to-material scatter is the rule, and a toothbrushing comparison of Beautifil Flow Plus X F00 with Beautifil II LS was not among the studies they included.

2.4 Conventional nanohybrid resins and gomers

Nanohybrid composites combine nano-sized particles with larger submicron or micron fillers. The dual size distribution is intended to increase packing density, improve polishability and raise filler load, which commonly exceeds 70–80 wt% ⁽¹¹⁾. Beautifil II LS belongs to this class. Its filler load is 68 vol%. The resin contains Bis-GMA, UDMA, Bis-EMA, PEGMA and TEGDMA, and the filler includes multifunctional glass and S-PRG particles ⁽¹⁹⁾.

S-PRG filler is a three-layered particle: a fluoro-boro-aluminosilicate glass core, a pre-reacted glass-ionomer hydrogel phase and an outer silica coating. It releases fluoride, strontium, borate, sodium, silicate and aluminium ions and can be recharged with fluoride ⁽²³⁾. These ion-exchange properties are the basis of the giomer concept and are clinically attractive. They are not, however, a guarantee of surface stability. Yap and colleagues showed that chemical challenge can roughen composite surfaces by softening the matrix and exposing filler ⁽²¹⁾, and the same ion-exchange behaviour that buffers acid may leave a more irregular topography once the resin-rich layer has been brushed away. Larger and irregular fillers, once uncovered, produce a higher Ra than a uniform nanofilled injectable ⁽¹⁸⁾.

Clinical evidence for giomer pastes is longer than the laboratory record for Beautifil II LS. Gordan, Mondragon, Watson, Garvan and Mjör followed an earlier Beautifil restoration placed with a self-etch primer for eight years and reported acceptable retention with little secondary caries in the teeth available for recall ⁽²²⁾. That series establishes that S-PRG pastes have been used as definitive restoratives for many years. It does not describe the toothbrush roughness of Beautifil II LS.

Packable nanohybrids remain the clinical reference for posterior direct work because of their handling familiarity and their documented wear record ⁽¹¹⁾. Their viscosity makes them more difficult to adapt in a narrow box and more likely to incorporate voids. Injectable products were introduced to close that handling gap. Whether they do so without a penalty in toothbrush wear or surface roughness is the question that the present experiment was designed to answer.

2.5 Statement of the problem

Two-body wear of G-ænial Universal Injectable and Beautifil Flow Plus F00 is favourable in one recent laboratory study ⁽⁶⁾. Toothbrush roughness of G-ænial Universal Injectable is favourable against a different nanohybrid in another ⁽¹⁸⁾. Beautifil Flow Plus X F00 and Beautifil II LS have not been placed beside G-ænial Universal Injectable in a single protocol that records both gravimetric loss and Ra after 10 000 cycles at 2 N. That comparison is the subject of this thesis.

CHAPTER 3

MATERIALS AND METHODS

3.1 Study design, setting and sample

The investigation was a comparative in-vitro study of three commercially available resin composites. Disc specimens were prepared in the laboratory of the Faculty of Dentistry, Pharos University in Alexandria. The toothbrushing runs were carried out on a custom simulator in the Dental Biomaterials Department of the Faculty of Dentistry, Alexandria University.

Thirty-six discs, 10 mm in diameter and 1 mm thick, were prepared in a CAD/CAM Teflon mould designed in Blender (Blender Institute B.V., Amsterdam). The discs were allocated at random to three equal groups (n = 12) according to the restorative material:

1. Group I — G-ænial Universal Injectable.
2. Group II — Beautifil Flow Plus X F00.
3. Group III — Beautifil II LS.

The sample size was calculated in G*Power version 3.1.9.2 from the two-body wear comparison of Rajabi et al. ⁽⁶⁾, taking a power of 80 % ($\beta = 0.20$) and a two-sided significance level of 0.05. That calculation gave a minimum of ten specimens per group (thirty in all). Twelve were prepared in each group. The protocol provided that any disc lost in processing would be replaced so that the analysed number remained twelve; no disc was lost.

3.2 Materials

The materials used were:

1. G-ænial Universal Injectable, shade A2 — highly filled nanofilled injectable composite (GC Corporation, Tokyo, Japan).

2. Beautifil Flow Plus X F00, shade A2 — highly filled injectable giomer of zero-flow consistency (Shofu Inc., Kyoto, Japan).
3. Beautifil II LS, shade A2 — conventional low-shrinkage nanohybrid giomer (Shofu Inc., Kyoto, Japan).
4. Colgate Total toothpaste.
5. Distilled water.

The composition of the three resins, as stated by the manufacturers, is summarised in Table 3.1. The instruments were a RADWAG AS 220-R2 analytical balance (RADWAG Wagi Elektroniczne, Warsaw), a MarSurf PS10 contact profilometer (Mahr GmbH, Göttingen) and the custom toothbrushing simulator already named.

Table 3.1 Brand, manufacturer, matrix, filler and load of the resin composites

Composite	Type	Matrix	Filler	Load	Manufacturer
G-aenial Universal Injectable (A2)	Nanohybrid injectable	Dimethacrylate monomers	Barium glass, silica	50 vol%	GC, Tokyo
Beautifil Flow Plus X F00 (A2)	Nanohybrid injectable giomer	Bis-GMA, TEGDMA	Multifunctional glass and S-PRG filler	47 vol%	Shofu, Kyoto
Beautifil II LS (A2)	Nanohybrid conventional giomer	Bis-GMA, UDMA, Bis-EMA, PEGMA, TEGDMA	Multifunctional glass and S-PRG filler	68 vol%	Shofu, Kyoto

Colgate Total was selected as the dentifrice because it is a widely used daily paste of moderate abrasivity and because it has been employed as the low-RDA reference paste in previous toothbrushing studies of resin composites ⁽²⁵⁾.

3.3 Specimen preparation

Each material was placed in the Teflon mould as a single increment. A Mylar strip and a transparent glass slide were applied over the unset paste so that the test face was flat, of uniform thickness and free of an oxygen-inhibited layer. No rotary finishing or polishing was done. The surface that entered the test was therefore the as-cured film against Mylar, not a polished clinical finish.

Polymerisation was carried out with an LED curing unit (MiniS, Woodpecker, Guilin, China) for 20 seconds at 800 mW/cm², with the tip in contact with the glass slide. Three further 20-second exposures were given from different directions so that a disc 1 mm thick was polymerised through. Flash was removed with a no. 12 scalpel blade (Suzhou Kyuan Medical Co., Jiangsu, China). After the disc had been taken from the mould its diameter was checked with a digital caliper. The discs were identified by group and stored in distilled water in a closed glass container at 37 °C for 24 hours before the baseline measurements were made.

3.4 Baseline measurements

Each disc was lifted from the storage water and dried with a sterile disposable sponge. It was then weighed on the RADWAG AS 220-R2 balance, which records to 0.0001 g. Surface roughness was measured on the test face with the MarSurf PS10. Five traces were taken at random sites on each disc; the device software calculated Ra for each trace, and the value entered for analysis was the mean of those five traces. Ra was determined according to the definition in ISO 4287 ⁽¹⁶⁾.

3.5 Toothbrushing protocol

The discs were mounted in the custom toothbrushing simulator so that the whole of the test face was brushed under a vertical load of 2 N. That load lies within the ISO range of 0.5–2.5 N ⁽³⁾. A slurry of Colgate Total (relative dentine abrasivity 70) and distilled water was prepared at 250 g of toothpaste per litre of water, in accordance with ISO 11609 and

ISO/TR 14569-1^(3,14). Each disc received 10 000 cycles. That count is the conventional laboratory equivalent of one year of twice-daily brushing, following the estimate of Sexson and Phillips^(7,8). The slurry was replaced every 5 000 cycles. At the end of the run the discs were rinsed with a stream of compressed tap water and dried with a disposable sponge.

3.6 Post-test measurements and calculations

Mass and Ra were recorded again with the same instruments and the same sponge-drying routine as at baseline, so that a change in mass would not be confused with a change in surface water.

Percentage weight loss was calculated for each disc as

$$[(W_1 - W_2) / W_1] \times 100$$

where W_1 and W_2 are the masses before and after brushing. The group value reported in Table 4.1 is the mean of these per-disc ratios, not the ratio of the group-mean masses. The distinction is material in a group that contains an outlier.

Absolute loss in milligrams was calculated as $(W_1 - W_2) \times 1000$.

The change in roughness was calculated for each disc as

$$\Delta Ra (\mu m) = Ra \text{ after} - Ra \text{ before.}$$

3.7 Statistical analysis

Descriptive statistics are presented as mean $\pm$ standard deviation. Percentage weight loss was compared among the three groups by one-way analysis of variance. Ra after brushing was compared by the Kruskal–Wallis test, with pairwise comparisons between groups. The choice of a parametric test for weight loss and a non-parametric test for roughness follows the distributions of the two outcomes in the experimental records. The name of the pairwise procedure used after the Kruskal–Wallis test was not recorded; the result that is reported is that each injectable differed from Beautifil II LS and that the two injectables did not differ from each other. The level of significance was set at $\alpha = 0.05$. The

statistical software used for the original analysis was not named in the laboratory records and is therefore not stated here.

CHAPTER 4

RESULTS

Thirty-six discs were tested, twelve in each group. None was lost in processing. All discs completed 10 000 cycles at 2 N. Wear is reported as mass; roughness is reported as Ra.

4.1 Weight loss

Mean mass before and after brushing, and percentage loss, are given in Table 4.1 and illustrated in Figure 4.1.

Table 4.1 Mean weight before and after simulated toothbrushing and percentage weight loss of the tested composite resins (n = 12)

Composite resin	Weight before (g)	Weight after (g)	Weight loss (%)	p-value
	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	
Beautifil Flow Plus X F00	0.1602 $\pm$ 0.0074	0.1576 $\pm$ 0.0031	1.49 $\pm$ 3.14	0.329†
G-aenial Universal Injectable	0.1630 $\pm$ 0.0102	0.1619 $\pm$ 0.0106	0.65 $\pm$ 0.37	
Beautifil II LS	0.2313 $\pm$ 0.0154	0.2303 $\pm$ 0.0156	0.41 $\pm$ 0.25	

† One-way ANOVA. Values are mean $\pm$ standard deviation.

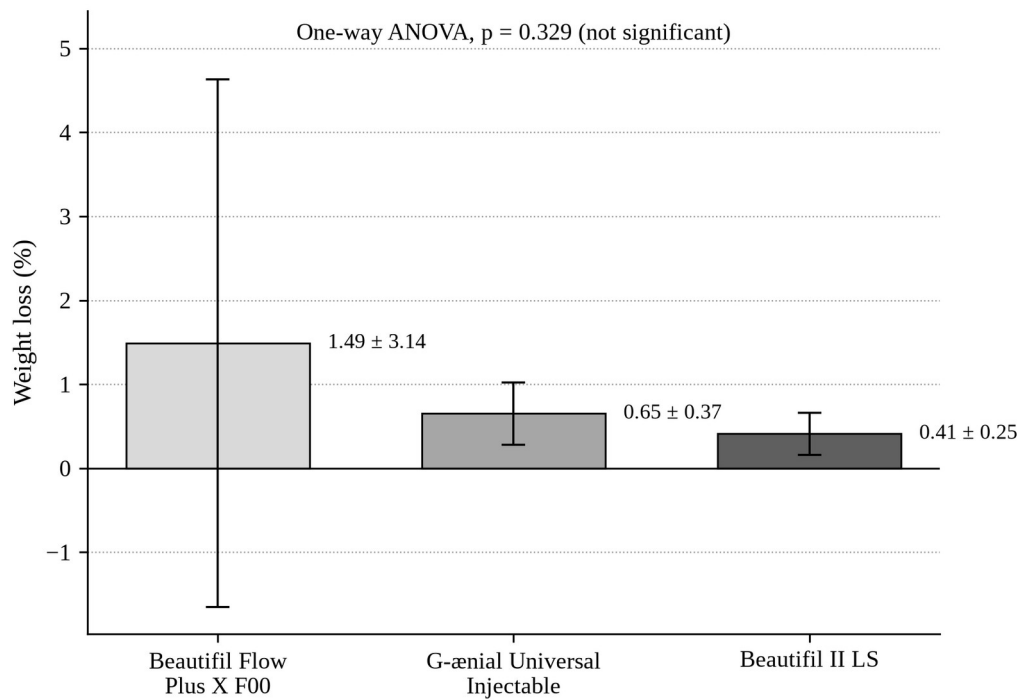


Figure 4.1 Mean percentage weight loss of the three composite resins after 10 000 toothbrushing cycles at 2 N. Error bars represent one standard deviation (n = 12 per group). In the Beautiful Flow Plus X F00 group the standard deviation exceeds the mean, so the lower error bar extends below zero; this reflects the dispersion introduced by a single specimen that lost approximately 11.4 % of its mass and does not imply a gain in weight. Differences among groups were not statistically significant (one-way ANOVA, $p = 0.329$).

Beautiful II LS discs were heavier at baseline (0.2313 ± 0.0154 g) than the two injectables (0.1602 ± 0.0074 g and 0.1630 ± 0.0102 g). This follows from filler load and density, not from a difference in disc size.

One-way ANOVA on percentage loss gave $p = 0.329$. The first null hypothesis is therefore retained. Numerically, Beautiful II LS lost the least mass (0.41 ± 0.25 %), followed by G-aenial Universal Injectable (0.65 ± 0.37 %) and Beautiful Flow Plus X F00 (1.49 ± 3.14 %). The standard deviation in the last group is an order of magnitude larger than in the other two.

Absolute loss is set out in Table 4.2 because percentage loss on a lighter disc can exaggerate a small mass change, and because the outlying specimen is easier to see in milligrams.

Table 4.2 Absolute weight loss after 10 000 brushing cycles (n = 12)

Composite resin	Absolute weight loss (mg)	Range (mg)
	Mean $\pm$ SD	
Beautifil Flow Plus X F00	2.58 $\pm$ 5.91	0.2 – 20.7
G-ænial Universal Injectable	1.07 $\pm$ 0.62	0.2 – 1.9
Beautifil II LS	0.95 $\pm$ 0.58	0.1 – 1.8

Values are mean $\pm$ standard deviation. The large standard deviation in Beautifil Flow Plus X F00 is attributable to one disc that lost approximately 11.4 % of its mass (20.7 mg). The remaining discs in that group, and all discs in the other two groups, lost between 0.1 mg and 1.9 mg.

The standard deviation of Beautifil Flow Plus X F00 mass fell from 7.4 mg before brushing to 3.1 mg after brushing. This is consistent with the heaviest disc in that group shedding 20.7 mg and landing near the group mean.

4.2 Surface roughness

Mean Ra before and after brushing, and Δ Ra, are given in Table 4.3 and illustrated in Figures 4.2 and 4.3.

Table 4.3 Mean surface roughness (Ra) before and after simulated toothbrushing and the change in roughness (Δ Ra) of the tested composite resins (n = 12)

Composite resin	Ra before (μ m) Mean $\pm$ SD	Ra after (μ m) Mean $\pm$ SD	Δ Ra (μ m) Mean $\pm$ SD	p-value
Beautifil Flow Plus X F00	0.038 $\pm$ 0.011	0.104 $\pm$ 0.017	0.066 $\pm$ 0.021	< 0.001‡
G-ænial Universal Injectable	0.054 $\pm$ 0.010	0.100 $\pm$ 0.023	0.046 $\pm$ 0.023	

Composite resin	Ra before (μm) Mean $\pm$ SD	Ra after (μm) Mean $\pm$ SD	ΔRa (μm) Mean $\pm$ SD	p-value
Beautifil II LS	0.119 ± 0.054	0.194 ± 0.050	0.075 ± 0.066	

‡ Kruskal–Wallis test. Pairwise comparisons: each injectable versus Beautifil II LS, $p < 0.001$; the two injectables, not significant. Values are mean $\pm$ standard deviation.

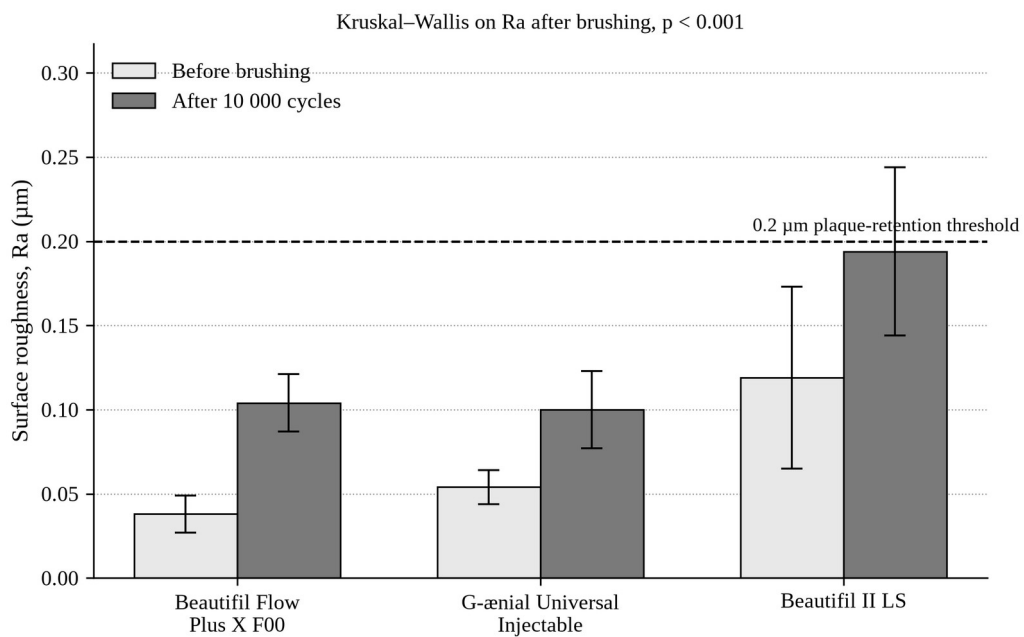


Figure 4.2 Mean surface roughness (Ra) of the three composite resins before and after 10 000 toothbrushing cycles at 2 N. Error bars represent one standard deviation ($n = 12$ per group). The broken line marks the $0.2 \mu\text{m}$ threshold above which plaque retention increases. Ra after brushing differed significantly among groups (Kruskal–Wallis, $p < 0.001$).

At baseline, on the as-cured Mylar face, Beautifil II LS was already the roughest ($0.119 \pm 0.054 \mu\text{m}$). Beautifil Flow Plus X F00 was the smoothest ($0.038 \pm 0.011 \mu\text{m}$). G-aenial Universal Injectable lay between them ($0.054 \pm 0.010 \mu\text{m}$). The order after brushing is therefore not a new ranking created by the toothbrush. It is the same order, carried forward.

Every group became rougher. Mean ΔRa was $0.066 \pm 0.021 \mu\text{m}$ for Beautifil Flow Plus X F00, $0.046 \pm 0.023 \mu\text{m}$ for G-aenial Universal Injectable and $0.075 \pm 0.066 \mu\text{m}$ for Beautifil II LS (Figure 4.3). After 10 000 cycles the means were $0.104 \pm 0.017 \mu\text{m}$, $0.100 \pm 0.023 \mu\text{m}$ and $0.194 \pm 0.050 \mu\text{m}$ respectively. Relative to its own baseline, Beautifil Flow

Plus X F00 rose by a factor of about 2.7, G-ænial Universal Injectable by about 1.9 and Beautifil II LS by about 1.6. The injectable that started smoothest therefore changed the most in relative terms, but it still finished beside the other injectable and well below the nanohybrid.

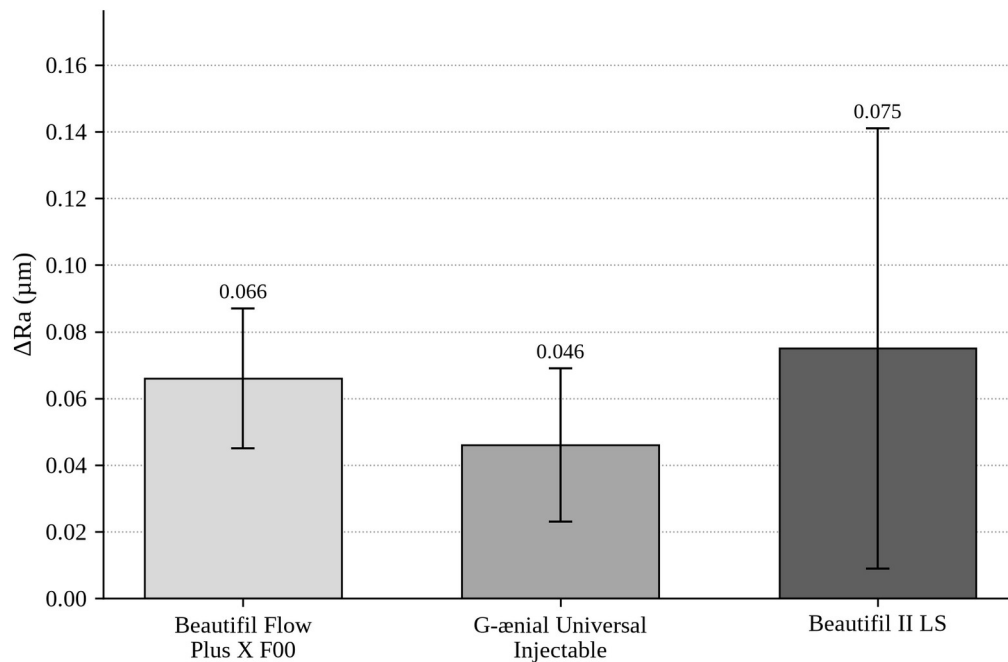


Figure 4.3 Mean change in surface roughness ($\Delta Ra = Ra$ after – Ra before) of the three composite resins after 10 000 toothbrushing cycles at 2 N. Error bars represent one standard deviation (n = 12 per group).

The Kruskal–Wallis test on Ra after brushing was significant ($p < 0.001$). The second null hypothesis is therefore rejected. Pairwise comparisons separated Beautifil II LS from each injectable resin but did not separate the two injectables from each other.

Relative to the $0.2 \mu m$ plaque-retention threshold ⁽¹⁾, both injectable means remained below it. The Beautifil II LS mean was $0.194 \mu m$, on the threshold. Its standard deviation of $0.050 \mu m$ indicates that some discs in that group crossed $0.2 \mu m$, which is visible as the upper error bar in Figure 4.2. All three means remained well below the $0.50 \mu m$ tongue-detection threshold ⁽²⁾.

4.3 Summary of the findings

Percentage weight loss did not differ significantly among the three resins. Absolute loss in the two well-behaved groups, and in eleven of the twelve Beautifil Flow Plus X F00 discs, was of the order of 1 mg. One disc in that group was an exception. After the same abrasion both injectable resins were significantly smoother than Beautifil II LS and did not differ from each other.

CHAPTER 5

DISCUSSION

The first null hypothesis is retained and the second is rejected. After 10 000 toothbrushing cycles at 2 N the three resins could not be distinguished on percentage weight loss, but they finished with significantly different surface roughness. The discussion that follows treats these two outcomes separately, because they describe different aspects of surface degradation and because they do not point to the same clinical conclusion.

5.1 Interpretation of the weight-loss findings

A non-significant analysis of variance ($p = 0.329$) is not evidence that the three materials wear equally. It is a failure to reject the hypothesis of equality under the conditions of this test. The mean percentage losses — 1.49 ± 3.14 % for Beautifil Flow Plus X F00, 0.65 ± 0.37 % for G-ænial Universal Injectable and 0.41 ± 0.25 % for Beautifil II LS — lie in the same numerical range once the outlying specimen is recognised. That specimen, a single Beautifil Flow Plus X F00 disc that lost approximately 11.4 % of its mass (20.7 mg), inflates both the group mean and, more importantly, the residual variance. With the variance enlarged, the power of the ANOVA to detect a genuine difference among the remaining discs is reduced. The other eleven discs in that group, and all twenty-four discs in the other two groups, lost between 0.1 mg and 1.9 mg. Viewed in that way, the three materials behaved similarly.

Beautifil II LS discs were heavier at baseline (0.2313 ± 0.0154 g) than the two injectables (approximately 0.16 g). The difference is a consequence of filler load and density, not of disc dimensions. Percentage loss on a heavier disc is a smaller number for the same milligrams removed, which is why absolute loss is the fairer comparison between groups: 0.95 ± 0.58 mg for Beautifil II LS against 1.07 ± 0.62 mg for G-ænial Universal

Injectable. Those two means are close. The nanofilled injectable, despite a lower filler weight fraction than the nanohybrid, did not lose more mass under this slurry.

The finding is consistent, in direction if not in method, with the two-body data of Rajabi et al., who reported lower volumetric wear for G-ænial Universal Injectable and Beautifil Flow Plus F00 than for a nanohybrid paste ⁽⁶⁾. Their test used 200 000 chewing cycles, a 49 N load and a steatite antagonist, without toothpaste, and they measured volume rather than mass. Beautifil Flow Plus F00 is, in addition, a different product from Beautifil Flow Plus X F00. The most that can be said is that highly filled injectables are not automatically the weaker partner when wear is measured in the laboratory. Condon and Ferracane's observation that wear rises once filler volume falls below 48 vol% is also relevant: G-ænial Universal Injectable is listed at 50 vol%, just above that step ⁽¹²⁾.

The origin of the 20.7 mg loss is not known. A subsurface void, an incompletely polymerised layer or an error in weighing would each produce an isolated high value. The specimen was retained and is identified in Table 4.2, because silent exclusion of an inconvenient point is not acceptable. Any subsequent analysis that omits it should present both the complete and the restricted data sets.

5.2 Interpretation of the roughness findings

The roughness result is the clearer of the two outcomes. Beautifil II LS was the roughest material before brushing ($0.119 \pm 0.054 \mu\text{m}$) and remained the roughest after it ($0.194 \pm 0.050 \mu\text{m}$). Its mean ΔRa was also the largest ($0.075 \pm 0.066 \mu\text{m}$). G-ænial Universal Injectable changed the least ($0.046 \pm 0.023 \mu\text{m}$). Beautifil Flow Plus X F00 started with the lowest baseline Ra ($0.038 \pm 0.011 \mu\text{m}$) and finished statistically indistinguishable from G-ænial Universal Injectable ($0.104 \pm 0.017 \mu\text{m}$ versus $0.100 \pm 0.023 \mu\text{m}$). The Kruskal–Wallis test on Ra after brushing was significant ($p < 0.001$), and the pairwise comparisons separated the nanohybrid from each injectable but not the two injectables from each other.

This order is the order predicted by filler geometry. A nanofilled injectable whose particles are of the order of 150 nm wears more uniformly, because the particles are small enough to be lost with the matrix rather than to stand proud of it ^(6,18,24). A nanohybrid that contains larger S-PRG and pre-polymerised particles loses the resin-rich surface first; the larger fillers are then uncovered or plucked, leaving a coarser profile ^(4,18). Hasan, Eltoukhy and Zaghloul recorded the same ranking after simulated brushing of G-aenial Universal Injectable against a nanofilled paste and a nanohybrid: the injectable was the smoothest, the nanohybrid the roughest, and the two nanofilled resins did not differ ⁽¹⁸⁾. Heintze et al. had already shown that hybrid and microhybrid pastes deteriorate more with load and time than microfilled resins ⁽⁴⁾, and da Costa et al. found that composites with smaller average fillers lost less gloss and gained less roughness when brushed with Colgate Total ⁽²⁵⁾. The present results sit within that literature rather than against it.

The 0.2 μm threshold of Bollen, Lambrechts and Quirynen remains the clinically useful reference ^(1,17). Both injectable means finished near 0.10 μm and therefore remain, on average, in the range where further polishing is not expected to reduce plaque. The Beautifil II LS mean of 0.194 μm lies on the threshold. The standard deviation of 0.050 μm indicates that a proportion of discs in that group exceeded 0.2 μm , a point that is visible as the upper error bar in Figure 4.2. None of the group means approached the 0.50 μm tongue-detection threshold of Jones et al. ⁽²⁾. Patients would not be expected to feel these surfaces after one laboratory year of this brushing model. The distinction that matters is biofilm risk, and it applies to the nanohybrid group.

The ΔRa standard deviation of Beautifil II LS (0.066 μm) is larger than both its baseline and its final Ra standard deviations. That pattern is possible only if the change in roughness is negatively correlated within discs — the roughest specimens becoming relatively smoother and the smoothest becoming rougher — and it is consistent with ΔRa having been calculated per disc before the group mean was formed.

5.3 Clinical implications

Within the limits of this protocol, the two injectable resins may be used on surfaces that are subject to toothbrush abrasion without a roughness penalty relative to Beautifil II LS. They do not, on these data, wear less. They finish smoother. A smoother surface after hygiene is an advantage on freely cleanable faces and at accessible margins, where plaque retention and stain are the practical concerns ^(1,17). The result does not support a claim that the same materials will resist occlusal contact wear as well as, or better than, a packable nanohybrid, because occlusal contact was not tested. The caution of Tzimas et al., that laboratory mechanical properties of highly filled flowables remain inferior to those of medium-viscosity composites and that use in extensive load-bearing cavities should be considered carefully, is therefore not displaced by the present findings ⁽²⁰⁾.

Beautifil II LS remains a reasonable packable control. Its higher baseline Ra is the as-cured Mylar face of a nanohybrid that contains larger S-PRG particles, not the face of a disc that has been taken through a polishing sequence. The two injectables, cured against the same strip, started smoother. Part of the after-brushing difference was therefore already present before the first stroke. The comparison after abrasion is nevertheless the comparison the study was designed to make: the three groups were treated identically and were compared as they stood at the end of the test.

A polished clinical restoration would not enter service at these baseline values. AlAli et al. started from polished nanohybrid surfaces of 0.14–0.22 μm and, after 10 000 cycles, were already rougher than at baseline; by 20 000 cycles their means had reached 0.41–0.49 μm ⁽²⁶⁾. The present injectable means after 10 000 cycles (approximately 0.10 μm) are lower than those polished-and-brushed figures. The comparison is not exact — the materials and the starting finish differ — but it is enough to say that the injectable surfaces in this study did not deteriorate into the range that AlAli recorded for polished nanohybrids under a longer run.

5.4 Strengths of the study

The three products are current commercial materials, and the Shofu injectable is the X-generation Flow Plus rather than the earlier F00 formulation used by Rajabi et al. ⁽⁶⁾. Twelve specimens were tested in each group rather than the ten indicated by the original power calculation. Mass and Ra were recorded on the same discs, so the two outcomes describe the same surfaces. The brushing load, the slurry ratio and the cycle count follow the approved protocol and the published ISO guidance ^(3,7,14). The statistical tests were chosen to match the distributions of the two outcomes. The outlying specimen was retained and identified rather than discarded.

5.5 Limitations of the study

The investigation is an in-vitro abrasion test. Saliva, pH cycling, enzymatic activity and occlusal load are absent, and the ranking obtained here may change when those factors are added ^(10,28). Gravimetric loss is not worn volume and is not worn anatomy ⁽¹⁵⁾. A disc that has spent 24 hours in water can also change mass by taking up or losing water; the same sponge-drying step was used before and after brushing in order to limit that error, but it cannot be excluded. A single dentifrice of moderate abrasivity was used, at 250 g per litre; a whitening paste of higher RDA might enlarge the roughness differences ^(8,25). Brush stiffness was not recorded. Scanning electron microscopy was not performed, so the inference that larger fillers are uncovered or plucked rests on the roughness numbers and on the published literature rather than on micrographs of these discs. Gloss was not measured. Ten thousand cycles represent one conventional laboratory year, not a year of any particular patient's hygiene. The test face was an as-cured Mylar surface, not a polished restoration. Disc geometry is not a Class II cavity, and the results should not be read as a prediction of wear at an occlusal contact or at a gingival margin. The statistical software and the name of the Kruskal–Wallis pairwise procedure were not recorded.

5.6 Concluding remarks of the discussion

After 10 000 strokes at 2 N the three resins could not be separated on percentage weight loss. They could be separated on Ra. Both injectable materials remained significantly smoother than Beautifil II LS and finished below the plaque-retention threshold; the nanohybrid finished on that threshold. That is the result of the study. It should not be enlarged into a general ranking of injectable versus nanohybrid composites outside the specific conditions of toothbrush abrasion.

CHAPTER 6

CONCLUSIONS AND RECOMMENDATIONS

6.1 Conclusions

Under the conditions of this in-vitro study:

1. Percentage weight loss after 10 000 toothbrushing cycles did not differ significantly among Beautifil Flow Plus X F00, G-aenial Universal Injectable and Beautifil II LS ($p = 0.329$).
2. Ra after the same abrasion did differ ($p < 0.001$). Both injectable resins were significantly smoother than Beautifil II LS. The two injectable resins did not differ from each other.
3. Mean Ra of both injectable resins remained below $0.2 \mu\text{m}$. Mean Ra of Beautifil II LS was $0.194 \mu\text{m}$.
4. The same order of roughness was already present on the as-cured Mylar face before brushing.
5. One Beautifil Flow Plus X F00 disc lost approximately 11.4 % of its mass and is the source of that group's large standard deviation.
6. The first null hypothesis is retained. The second is rejected.

6.2 Recommendations

Clinical practice. Where toothbrush abrasion and the smoothness of the surface after daily hygiene are the principal concerns, either injectable resin tested here may be considered alongside Beautifil II LS. The present data do not show that the injectables wear less; they show that they finish smoother. The result should not be used to justify the substitution of an injectable resin for a packable nanohybrid on occlusal contacts.

Further laboratory work. The mass analysis should be repeated with and without the outlying disc, and both calculations should be reported. Volumetric loss or profilometric

wear depth would complement the gravimetric outcome. Scanning electron microscopy of the worn surface would test the inference that larger fillers are uncovered in the nanohybrid. The same three materials should be finished and polished as they would be in the mouth, and then brushed, so that the ranking can be read against a clinical starting surface. Brush stiffness and the statistical software should be stated. A second dentifrice of higher relative dentine abrasivity, or a higher cycle count, would show whether the roughness difference persists under a more aggressive challenge.

Clinical research. Only a controlled clinical study can determine whether the difference between approximately 0.10 μm and 0.19 μm after a laboratory year is associated with a difference in plaque accumulation, staining or restoration survival.

CHAPTER 7

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